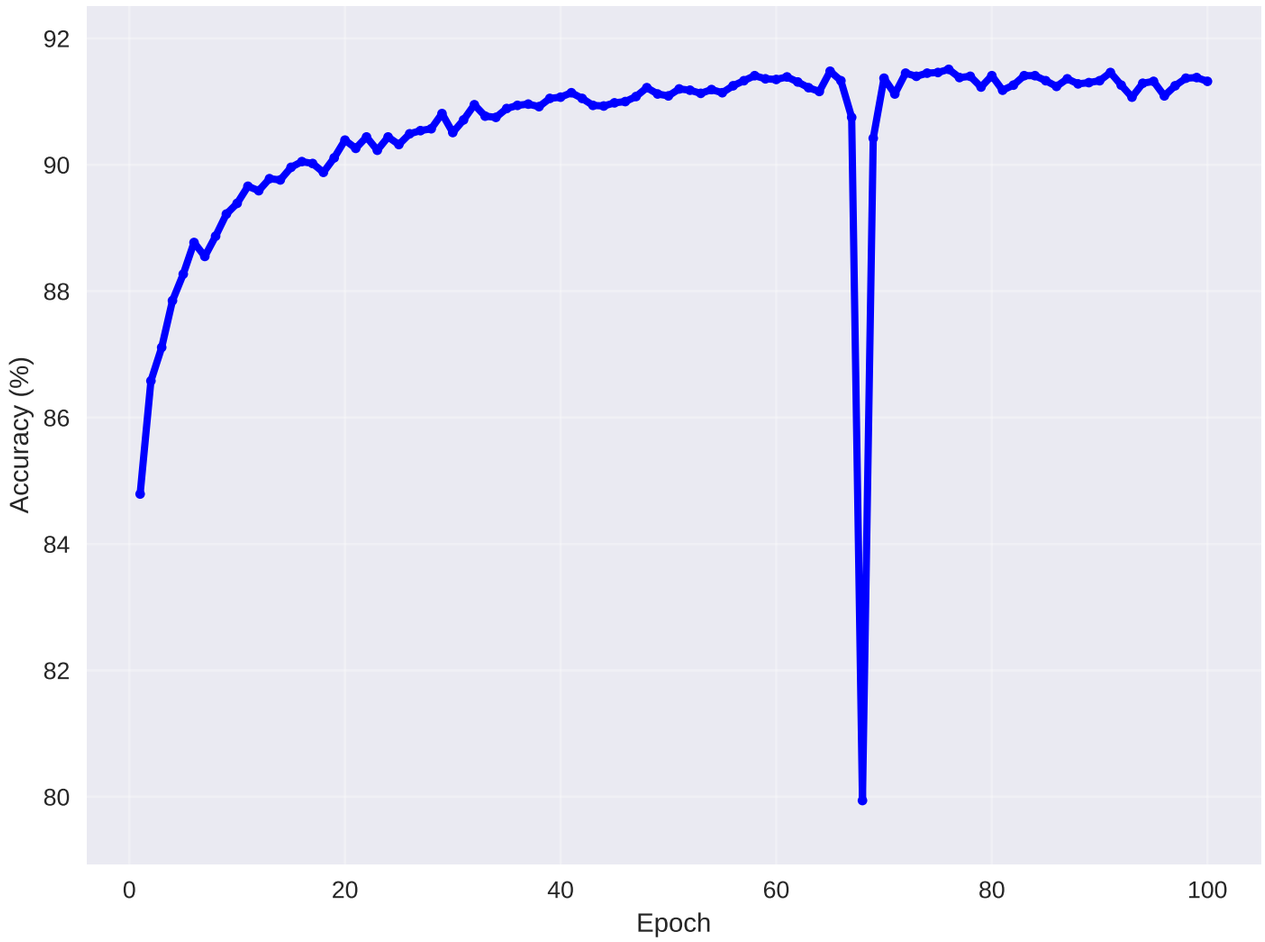
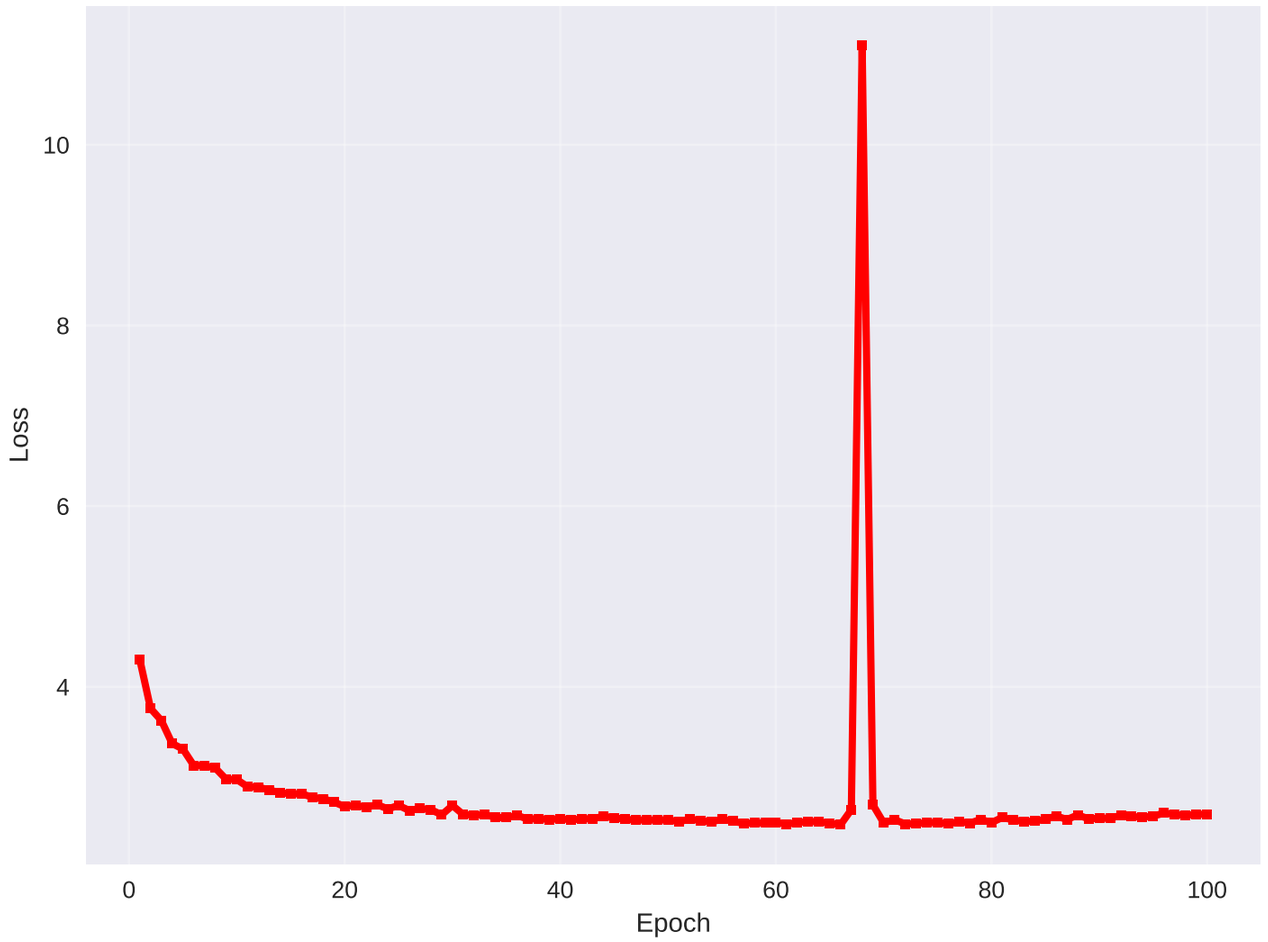
Federated Learning - Main Metrics Overview

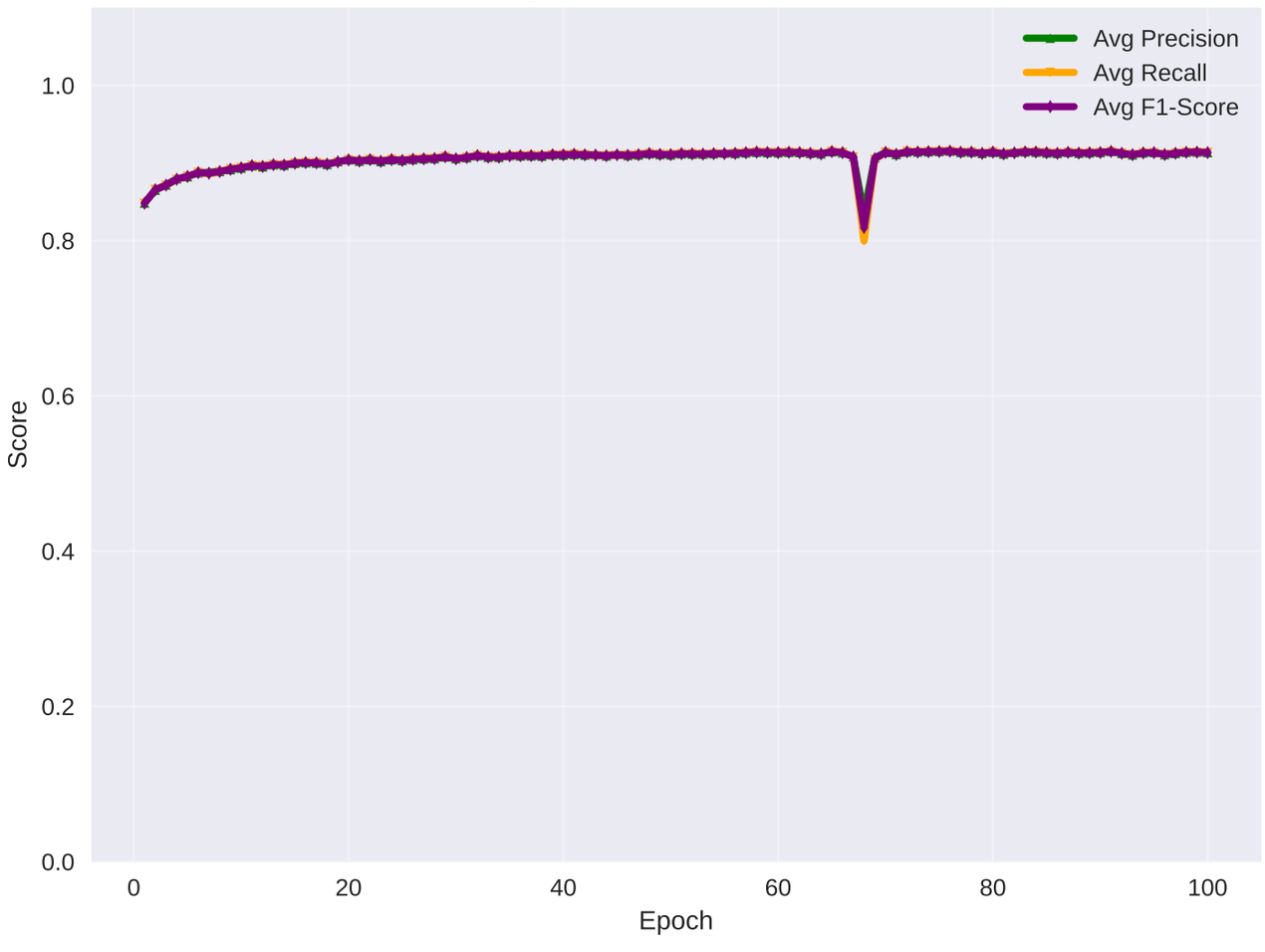
 Global Model Accuracy Evolution



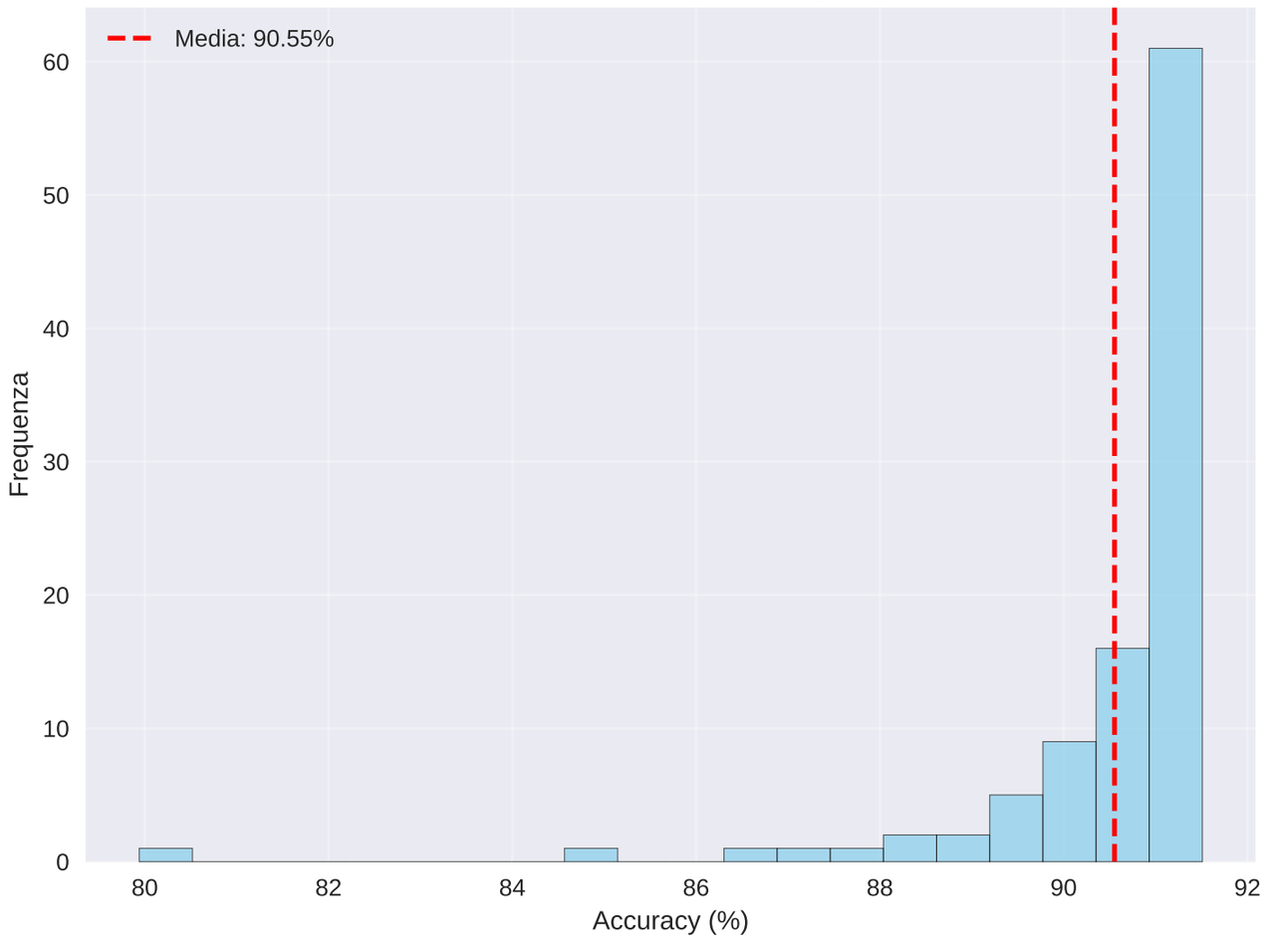
 Global Model Loss Evolution



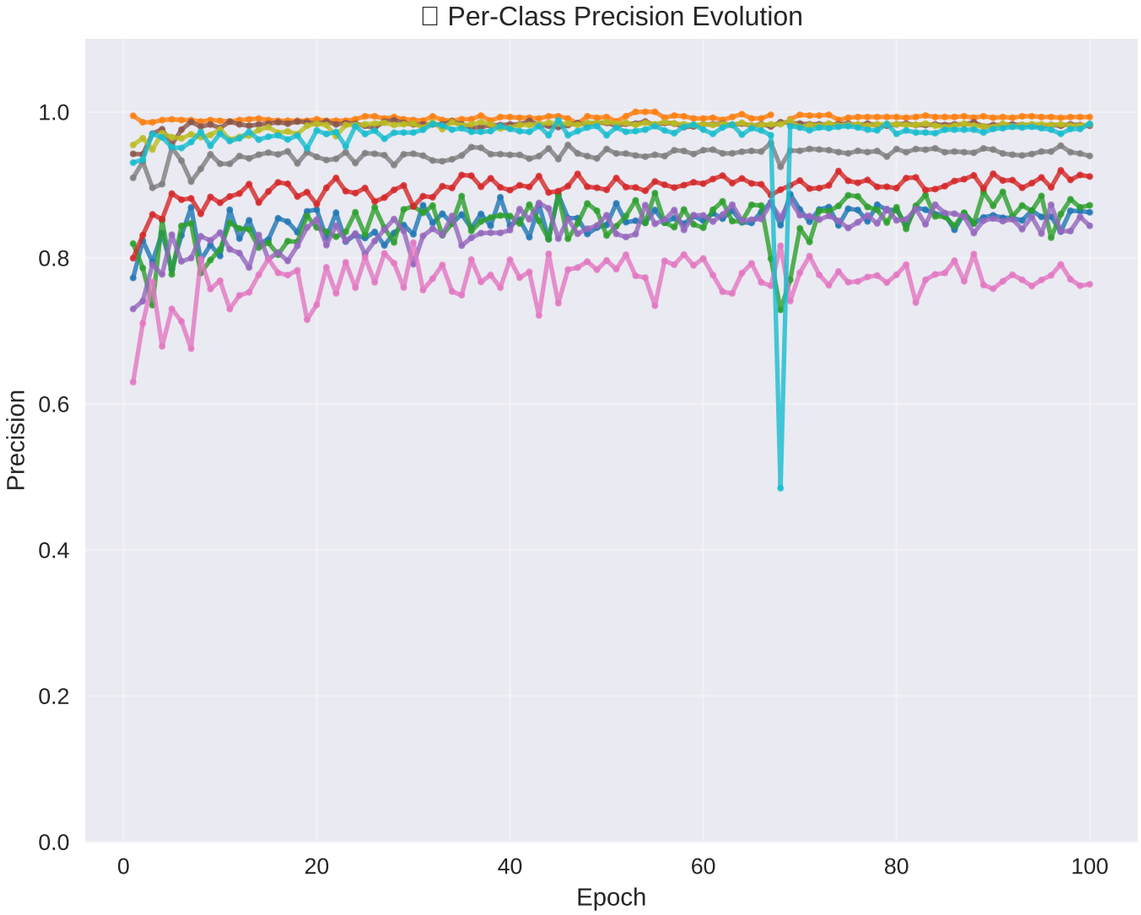
 Average Precision, Recall & F1-Score

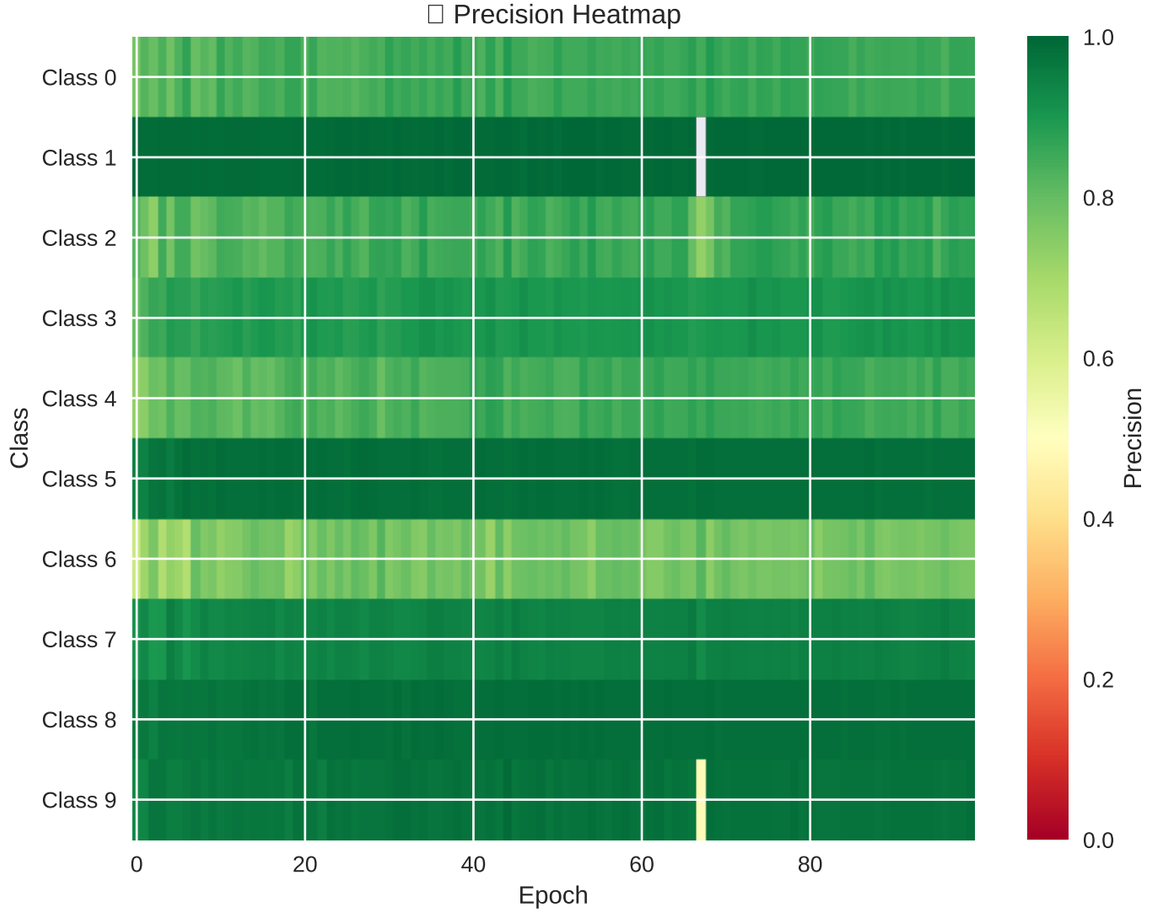


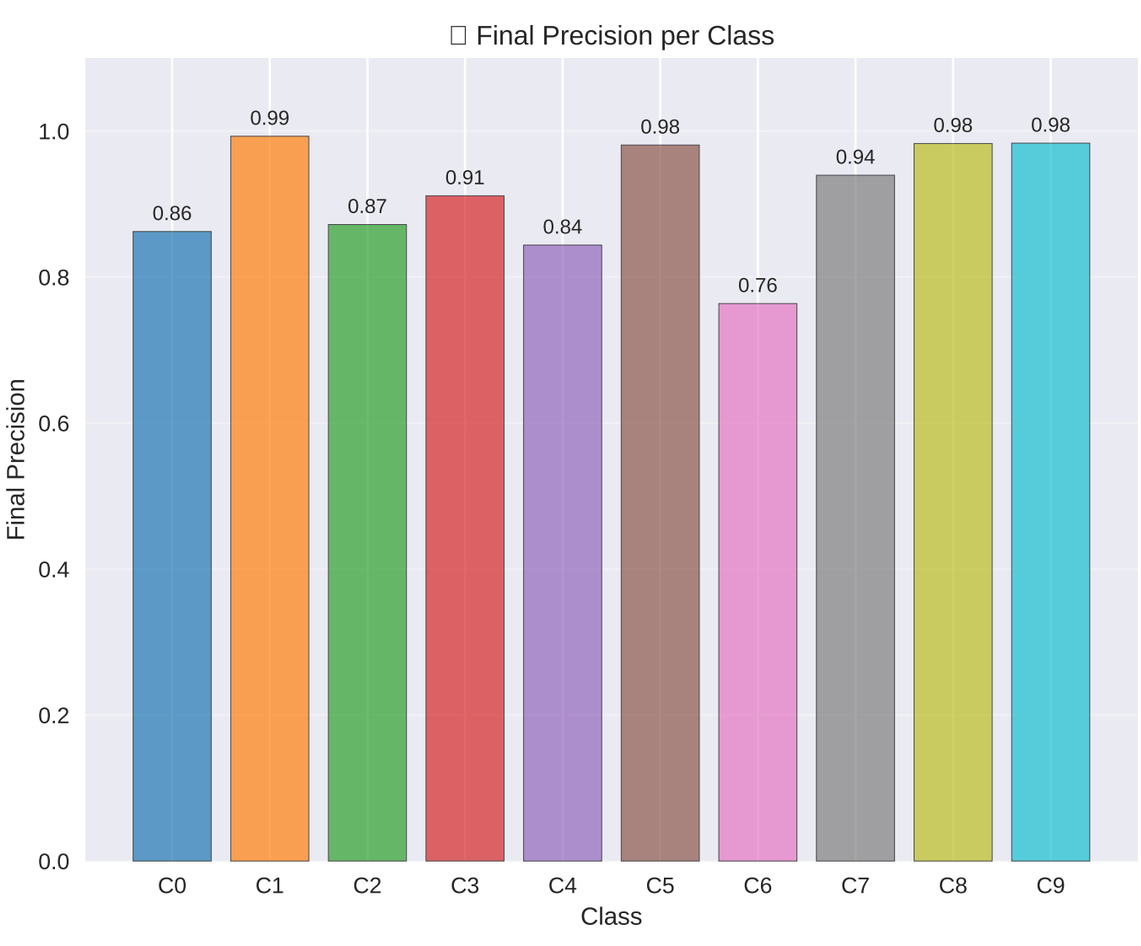
 Distribuzione Accuracy

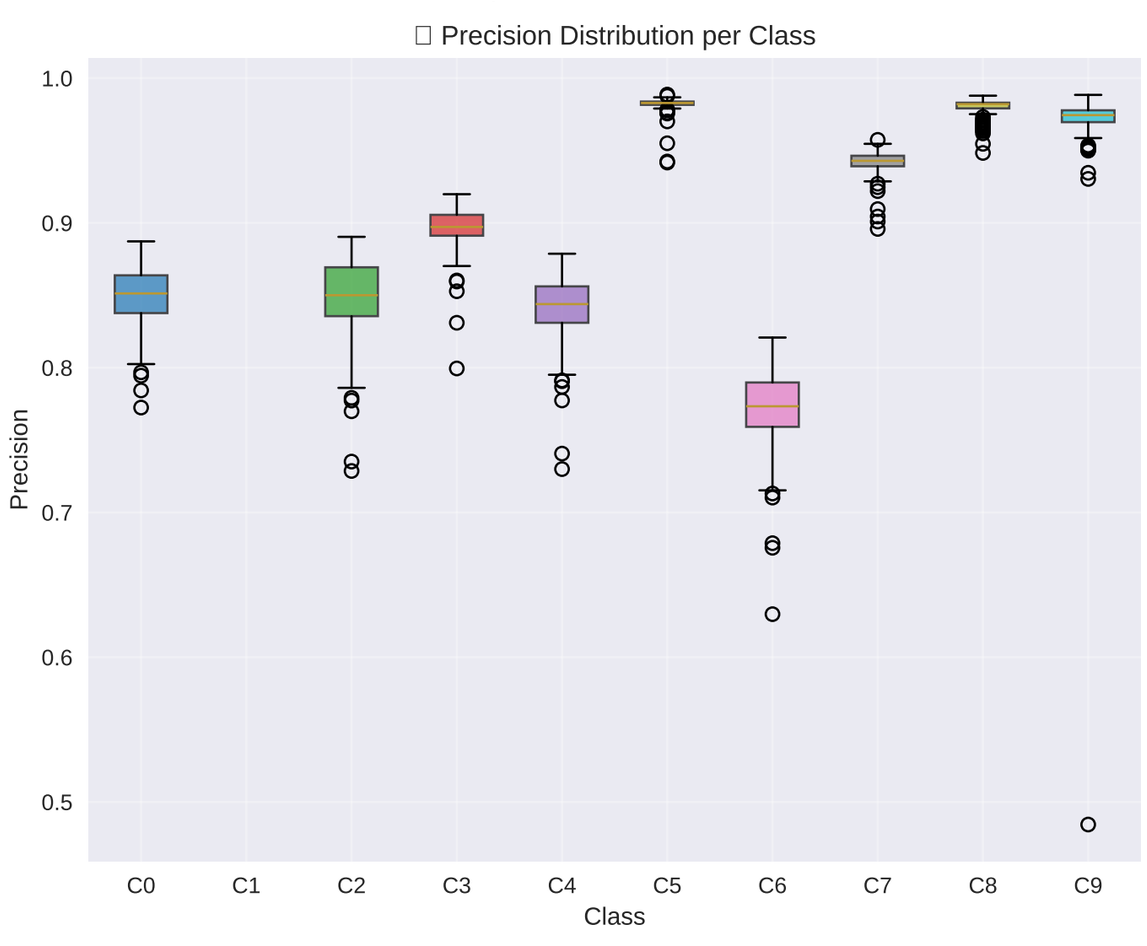


Per-Class Precision Analysis

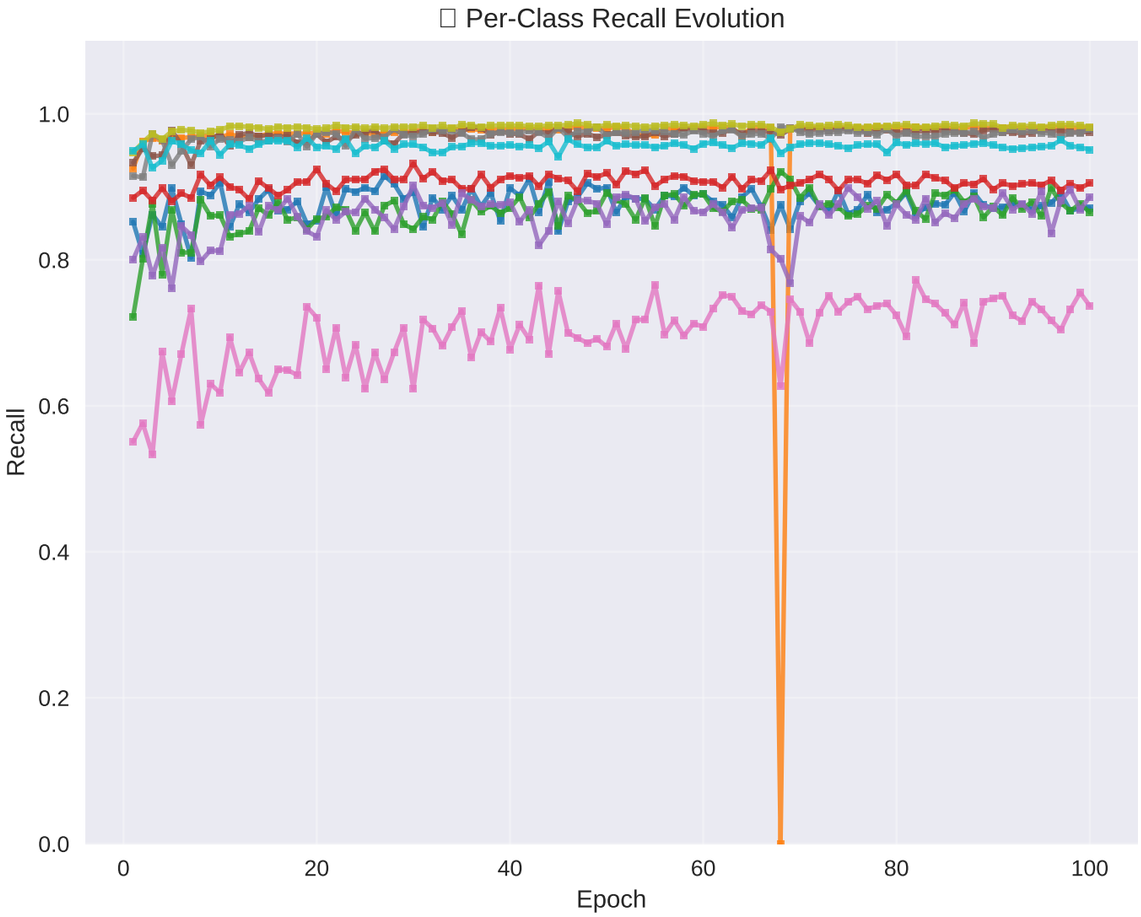


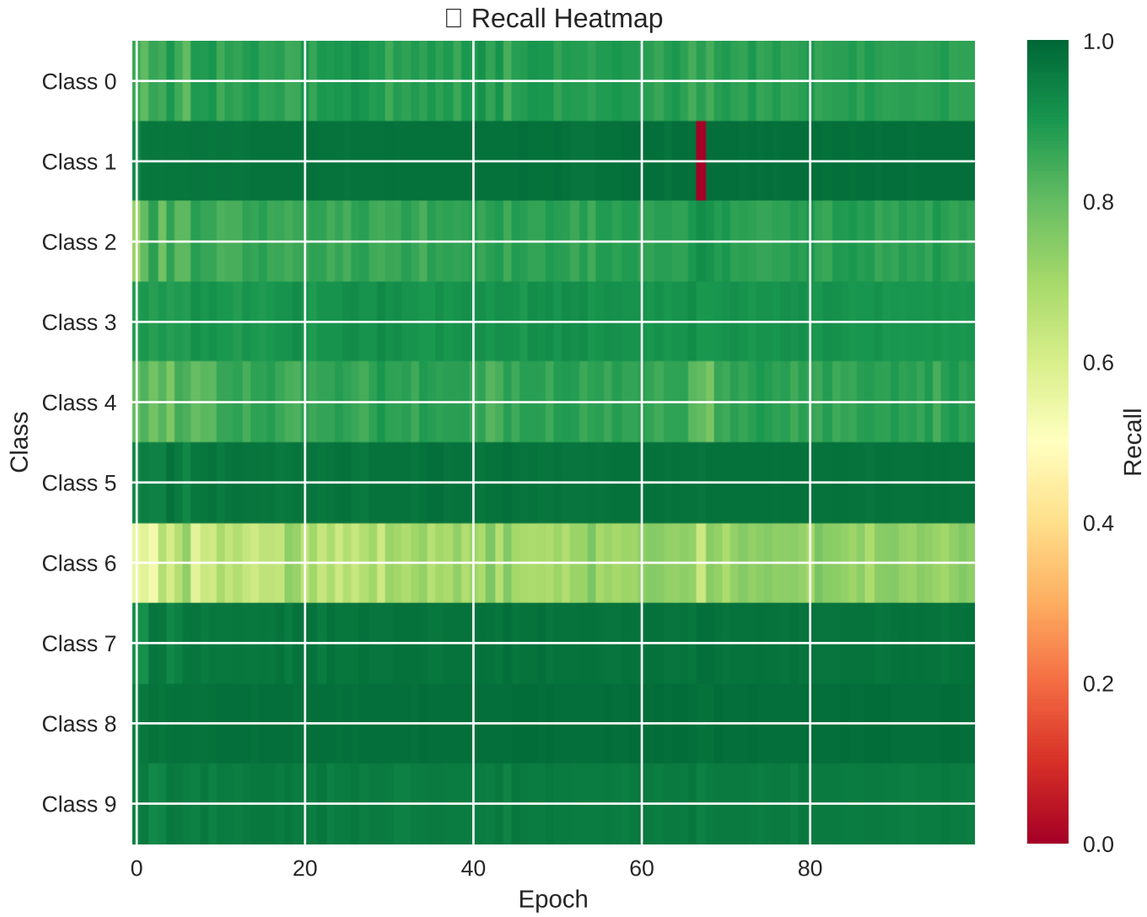


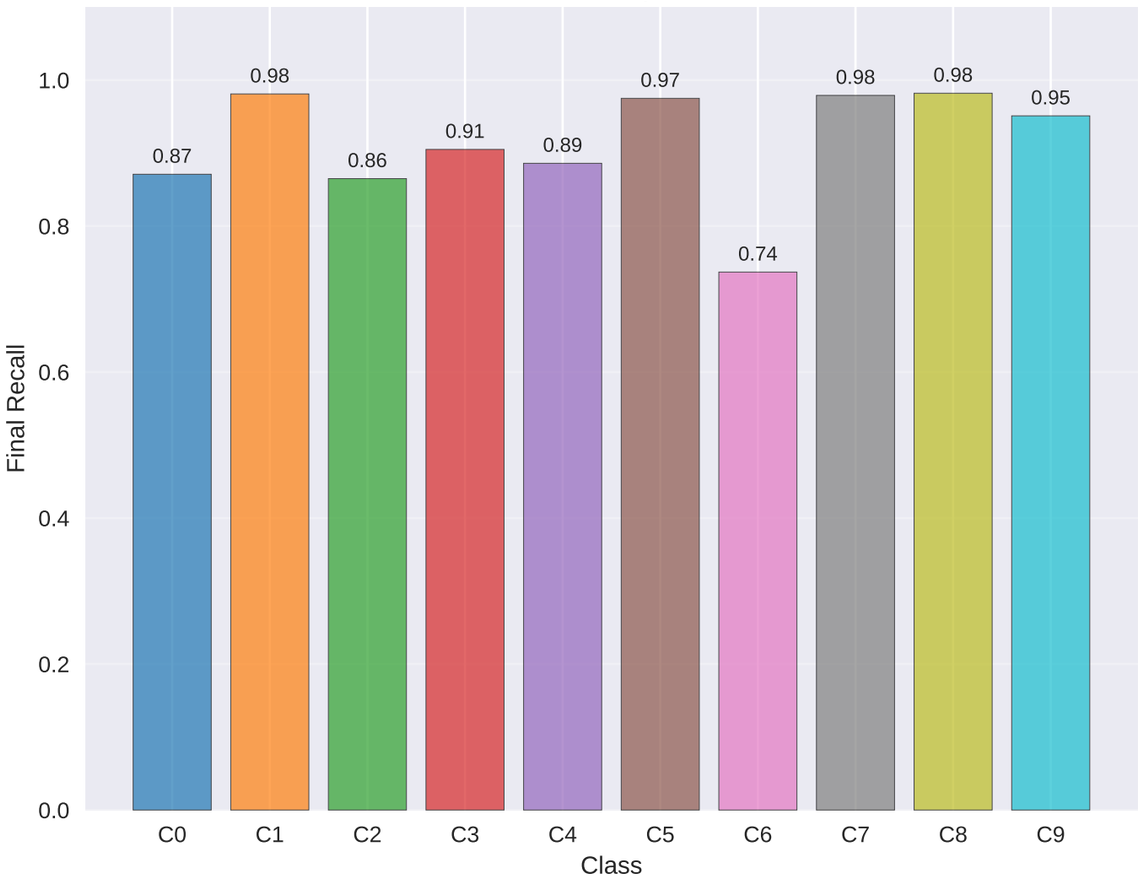


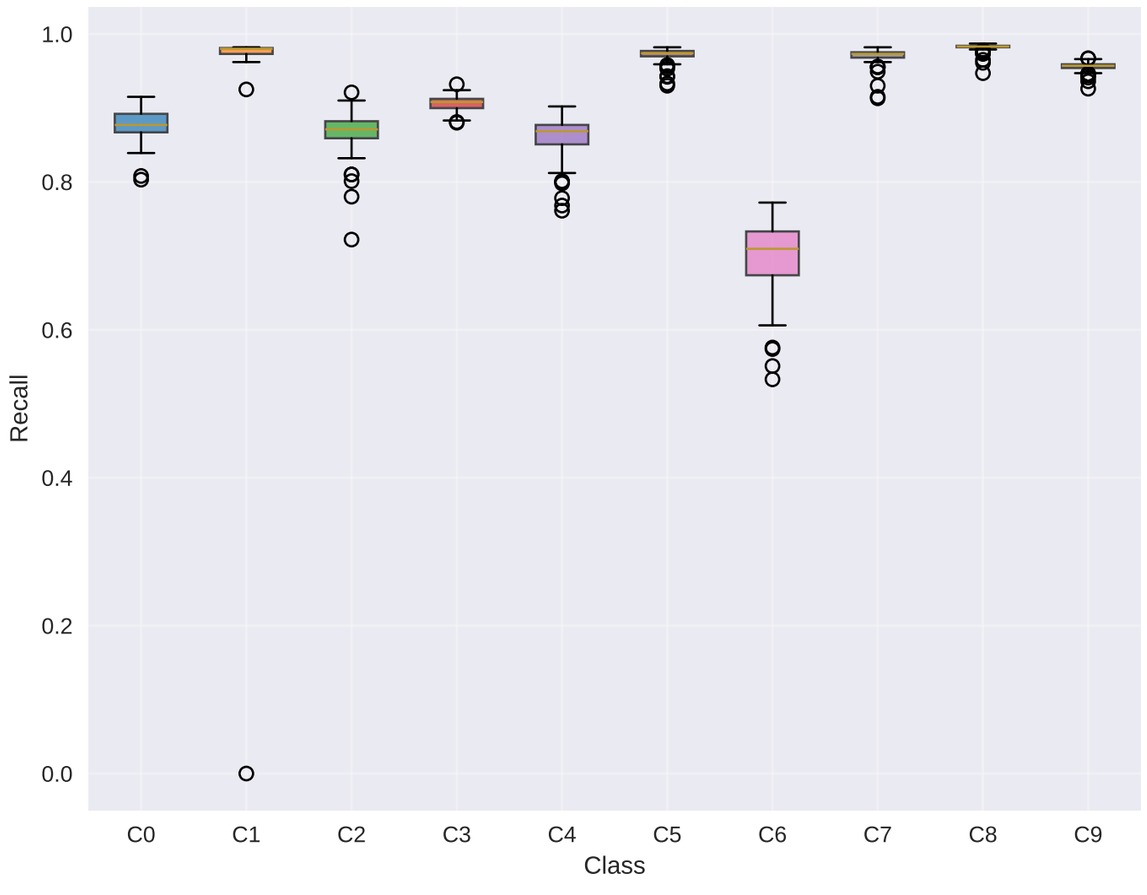


Per-Class Recall Analysis

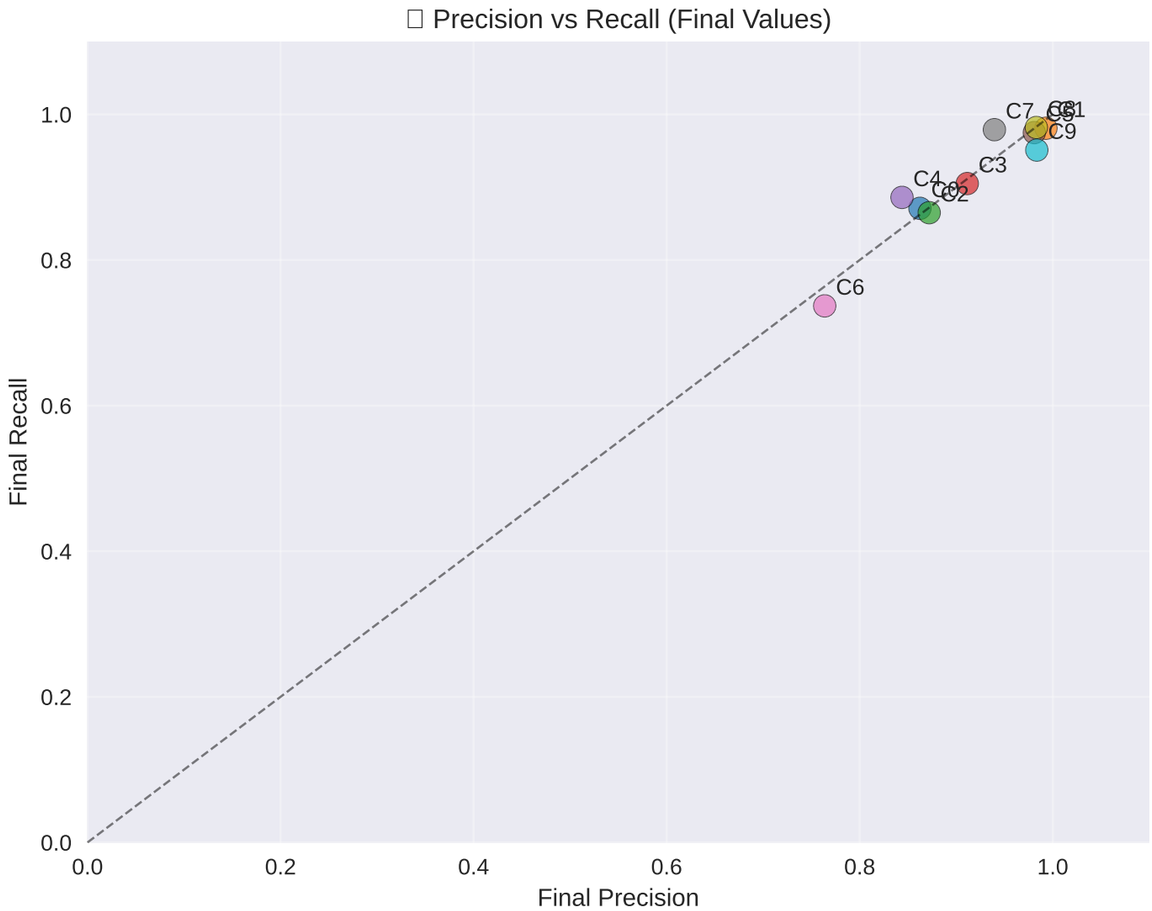


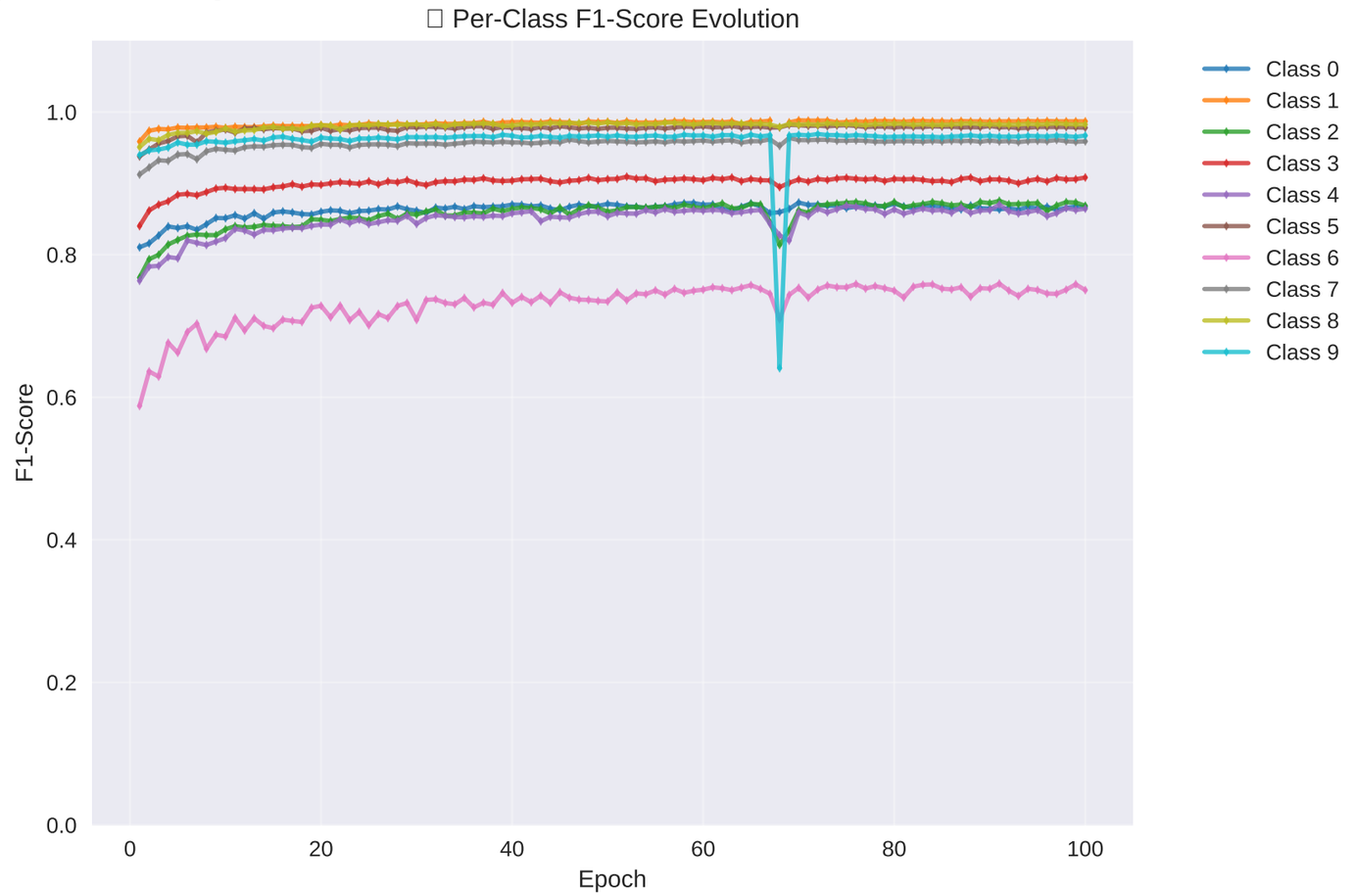


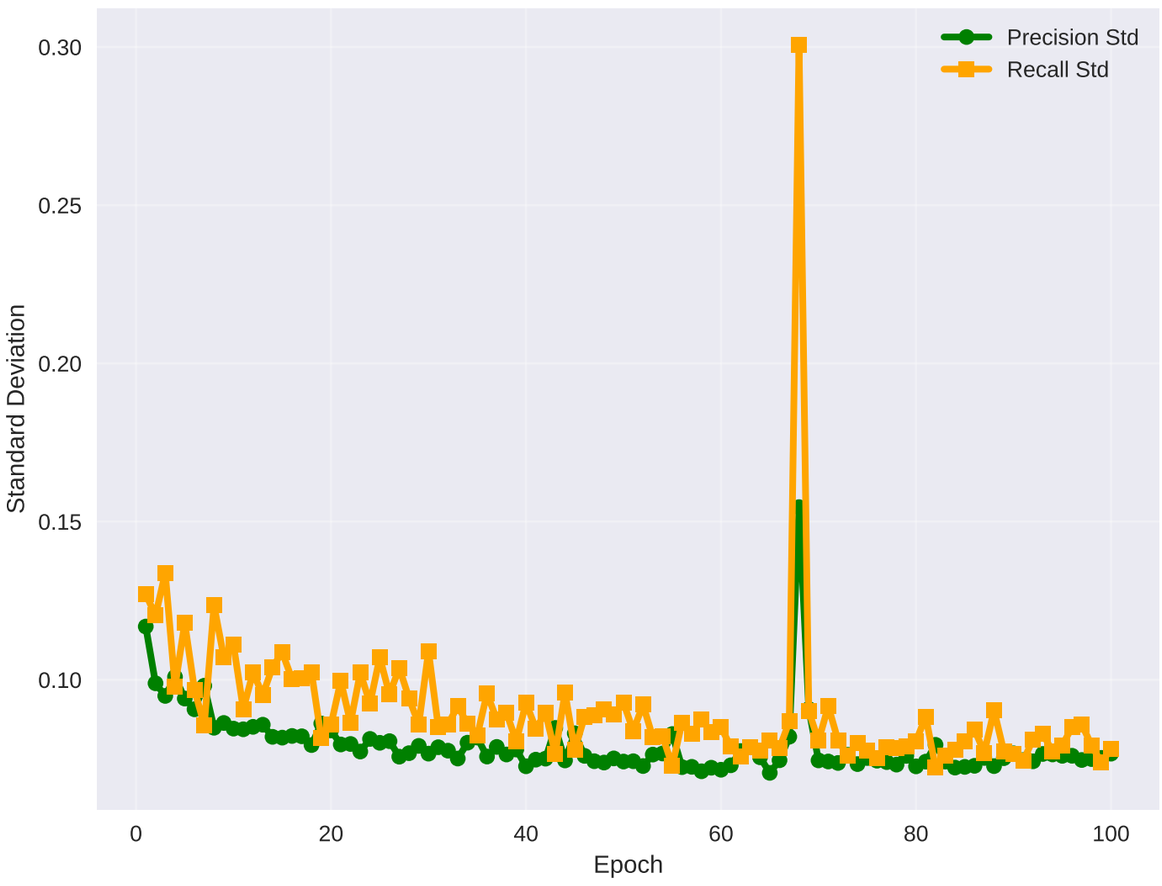


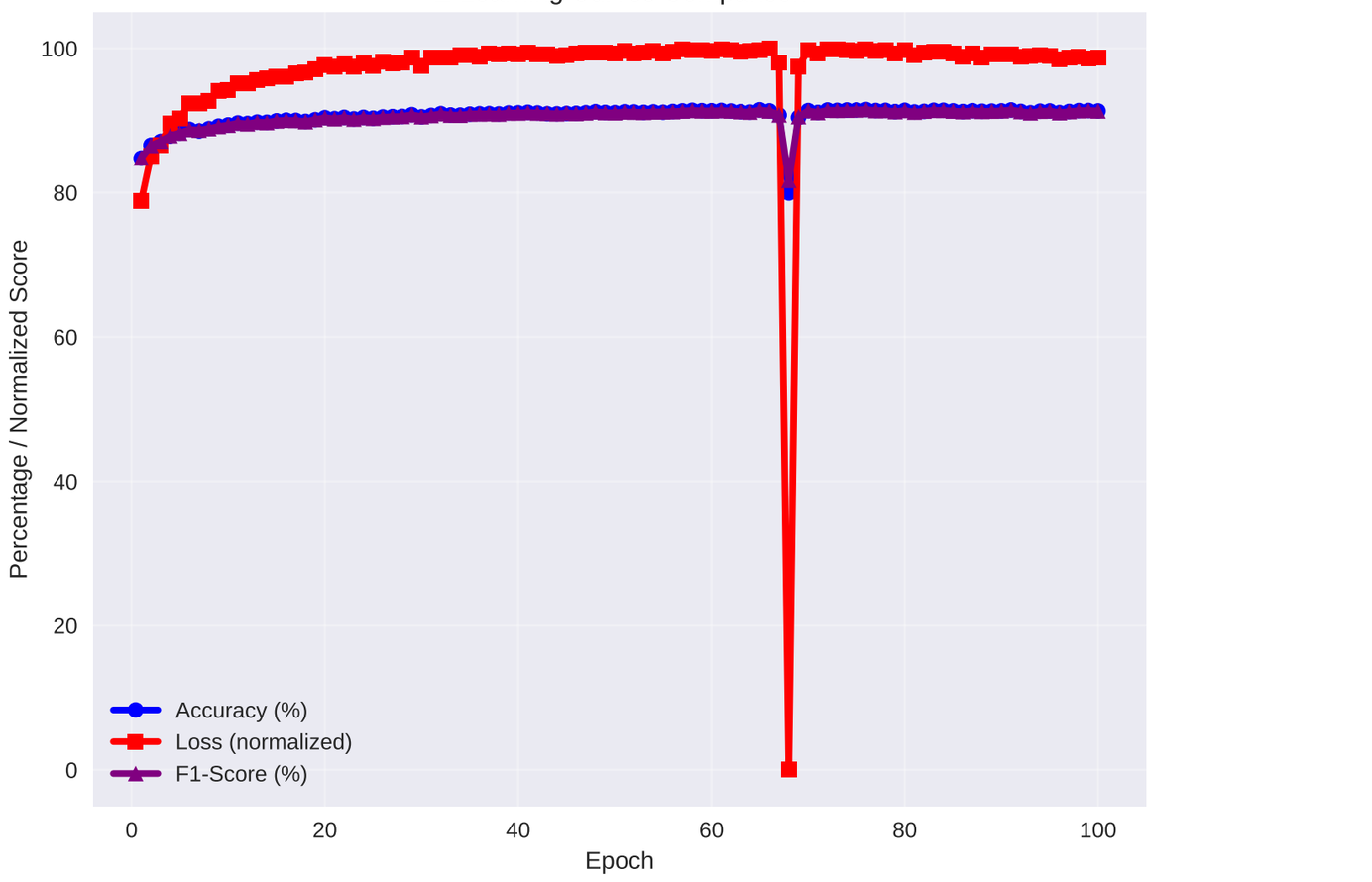


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-31 19:41:14
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.32%
• Loss: 2.5878
• Average Precision: 0.913
• Average Recall: 0.913
• Average F1-Score: 0.913

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.53%
• Riduzione Loss: +1.7115
• Stabilità Accuracy (CV): 0.0170

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.993)
• Peggiore Precision: Classe 6 (0.764)
• Migliore Recall: Classe 8 (0.982)
• Peggiore Recall: Classe 6 (0.737)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.862, R=0.871, F1=0.867
• Classe 1: P=0.993, R=0.981, F1=0.987
• Classe 2: P=0.872, R=0.865, F1=0.868
• Classe 3: P=0.911, R=0.905, F1=0.908
• Classe 4: P=0.844, R=0.886, F1=0.864
• Classe 5: P=0.981, R=0.975, F1=0.978
• Classe 6: P=0.764, R=0.737, F1=0.750
• Classe 7: P=0.940, R=0.979, F1=0.959
• Classe 8: P=0.983, R=0.982, F1=0.982
• Classe 9: P=0.983, R=0.951, F1=0.967

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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